Assignment 14

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1 Introduction

Image classification is one of the fundamental problems in computer vision. Traditional models like Fully Connected Feed-Forward Neural Networks (FCFNN) and Convolutional Neural Networks (CNN) have achieved remarkable success, especially CNNs due to their strong inductive biases toward spatial locality and translation invariance. However, the Vision Transformer (ViT) has recently emerged as a competitive alternative, leveraging self-attention to model long-range dependencies within an image.

This report explores the implementation and comparative analysis of a ViT-based image classifier on a subset of the ImageNet dataset containing 20 classes. The study also investigates the effects of different architectural hyperparameters of ViT, including the number of attention heads, patch sizes, and positional embedding strategies.

2 Problem Statements

1. Train a ViT-based classifier for 20 classes of the ImageNet dataset.
2. Compare the ViT-based classifier with a Fully Connected Feed-Forward Network (FCFNN) and a CNN-based classifier.
3. Analyze the effect of the number of attention heads on the performance of ViT.
4. Study the effect of patch size on ViT's performance.
5. Study the effect of positional embedding (none, fixed, learnable) on ViT's performance.
6. Discuss the challenges and difficulties of handling ViT compared to FCFNN and CNN.

3 Dataset Description

A balanced dataset containing 20 classes from the ImageNet dataset was used. Each class contains an equal number of samples.

Training samples: 8000
Testing samples: 2000

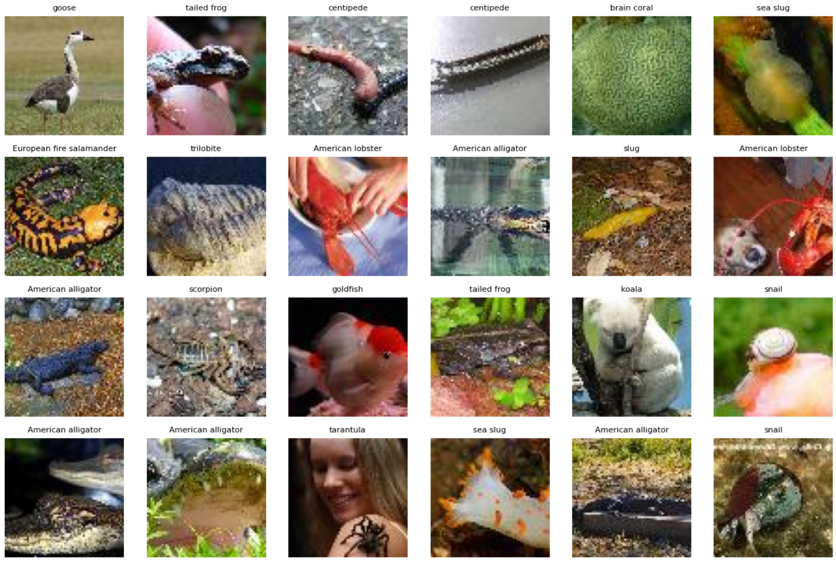

Figure 1: Visualization of sample data from 20 ImageNet classes

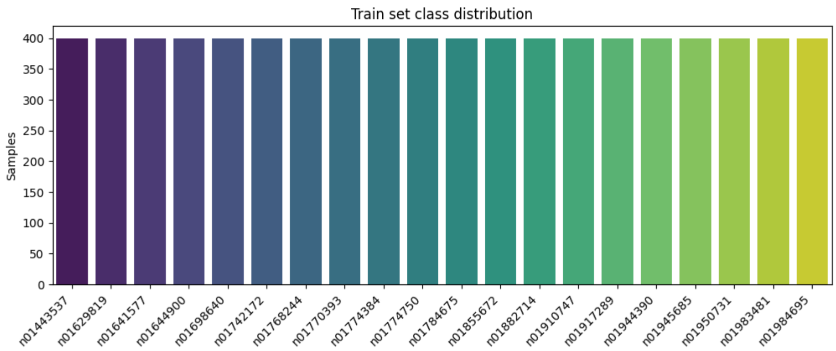

Figure 2: Train set class distribution across 20 classes

4 Model Architecture Overview

4.1 FCFNN

The FCFNN consists of multiple dense layers where each neuron is fully connected to all neurons in the previous layer. It lacks spatial inductive bias and is sensitive to input dimension, requiring flattened input images.

4.2 CNN

The CNN model uses convolutional layers with ReLU activation, followed by pooling and fully connected layers. It captures local spatial features effectively and provides translation equivariance, making it efficient for image-based tasks.

4.3 Vision Transformer (ViT)

The ViT model splits an image into non-overlapping patches, linearly embeds them, adds positional encodings, and processes them through transformer encoder blocks. Each block includes:

- Multi-Head Self-Attention (MHSA)
- Layer Normalization
- Multi-Layer Perceptron (MLP)

ViT treats images as sequences, enabling global context modeling but requiring large datasets or pretraining due to weak inductive biases.

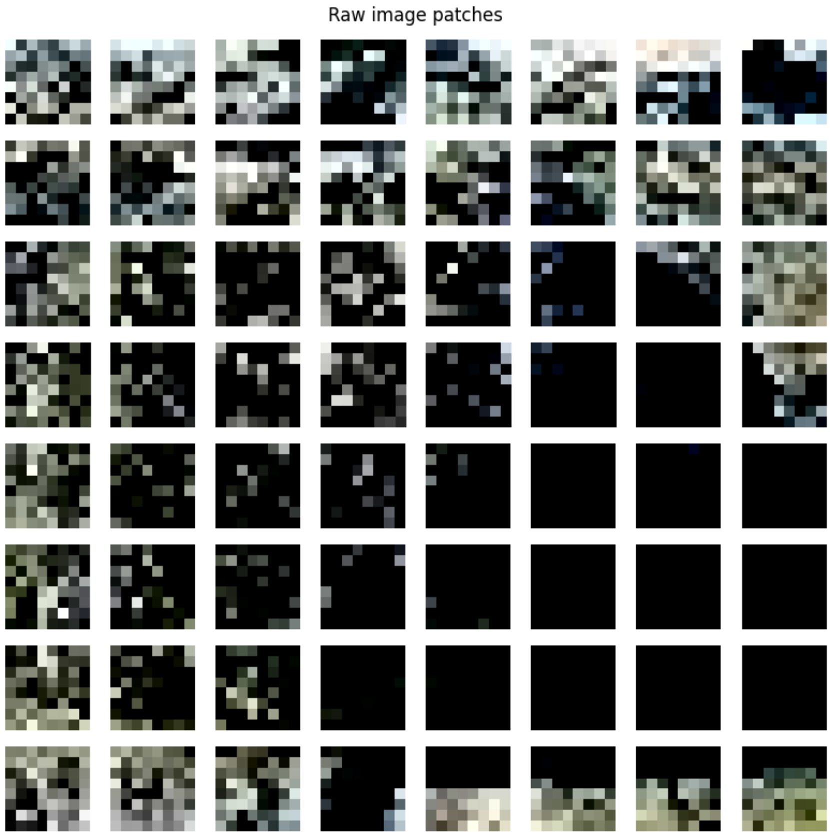

Figure 3: Raw image patches used as ViT input tokens

5 Experimental Results

5.1 Overall Model Comparison

After 10 epochs, the accuracy results are summarized in Table 1.

Table 1: Comparison of FCFNN, CNN, and ViT after 10 epochs

Model	Epochs	Train Accuracy (%)	Test Accuracy (%)
FCFNN	10	65.17	25.20
CNN	10	29.74	29.05
ViT	10	50.83	44.20

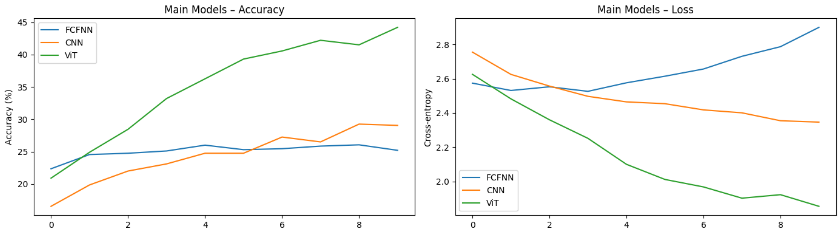

Figure 4: Model accuracy comparison of FCFNN, CNN, and ViT

Observation: While CNN typically performs well on large-scale image tasks, here it underperforms due to limited training epochs and small dataset size. ViT, despite slower convergence, achieves better generalization compared to CNN and FCFNN. FCFNN overfits quickly due to a lack of spatial feature learning.

5.2 Effect of Number of Heads

Table 2: Effect of attention heads on ViT performance

Heads	Epochs	Train Accuracy (%)	Test Accuracy (%)
1	10	44.35	40.05
2	10	49.30	42.00
4	10	51.06	43.95
8	10	54.58	42.75

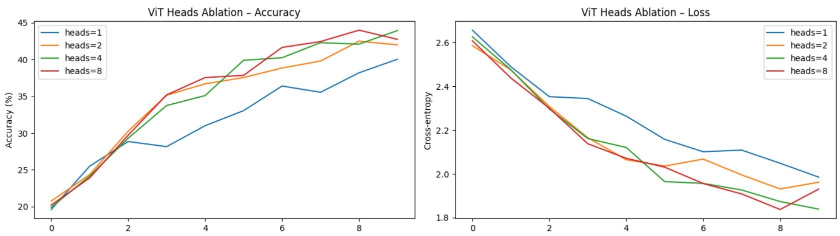

Figure 5: ViT Heads Ablation Accuracy

Observation: Increasing the number of heads improves model expressiveness up to a limit (4 heads performed best). Beyond that, overfitting and training instability slightly degrade test accuracy.

5.3 Effect of Patch Size

Table 3: Effect of patch size on ViT performance

Patch Size	Epochs	Train Accuracy (%)	Test Accuracy (%)
4	10	54.91	46.95
8	10	49.60	42.15
16	10	37.48	30.55

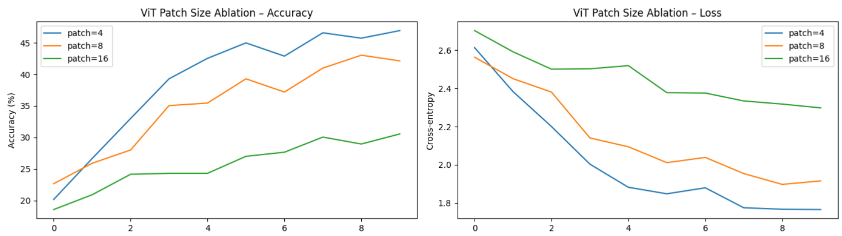

Figure 6: ViT Patch Size Ablation Accuracy

Observation: Smaller patches (4×4) preserve more fine-grained details and yield better accuracy. Larger patches (16×16) lose local structure, reducing performance.

5.4 Effect of Positional Embedding

Table 4: Effect of positional embedding choice on ViT performance

Positional Embedding	Epochs	Train Accuracy (%)	Test Accuracy (%)
None	10	46.79	39.75
Fixed	10	46.56	41.50
Learnable	10	50.10	43.60

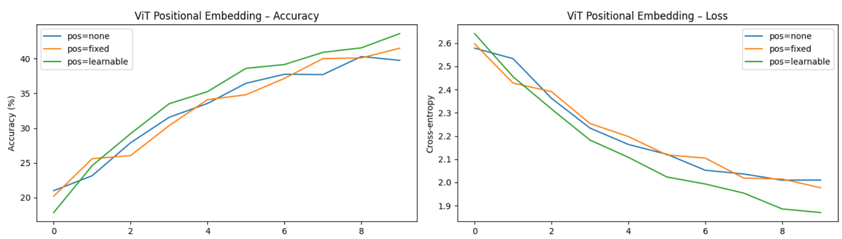

Figure 7: ViT Positional Embedding Accuracy

Observation: Learnable positional embeddings consistently outperform fixed or absent ones since they allow the model to adaptively learn spatial relationships during training.

6 Discussion and Difficulties of ViT

Aspect	FCFNN	CNN	ViT
Inductive bias	None	Strong (locality, translation equivariance)	Minimal (global context only)
Data efficiency	Poor	Good	Poor without pre-training
Convergence	Fast	Fast	Slow, heavy regularisation needed

Hyper-parameters	Few	Moderate	Many (heads, layers, patch, embed, dropout, learning rate schedule)
Memory	O(n)	O(hw)	O(n^2) with respect to patches
Interpretability	None	Grad-CAM visualization	Attention rollout possible

Source: Github